

- Q.1 Launch path with Clock - 2ns period, Insertion delay - 300ps, clk2q delay - 30ps, combo delay - 1.2ns, Tsetup with -10ps, Setup uncertainty -50ps, Capture Path with Clock - 3ns period, Insertion delay - 300 ps,
Will it meet setup and Hold ? If not why ? similarly try with different clock periods and insertion delays and check will you meet hold and setup ?.
- Q.2 After routing if you have 1000 unique setup violations? how we will fix? can we fix with tool? How?
- Q.3 How can you write script to fix 1000 setup unique violations?
- Q.4 Can you build CTS without SDC? If it is not why? Without SDC can we close the design?
- Q.5 Can you close the design without Clock tree (I'm not bothering timing)
- Q.6 In which stage SDC is required? Can you place design without SDC?
- Q.7 What will happen, when Victim is at constant high, Aggressor changing from logic 0 to logic 1.
- Q.8 If design meets 300MHz, does it meet 1GHz?
- Q.9 How we will balance the multiple clocks in a design?
- Q.10 Insertion delay will affect on frequency? How? What is the relation between Insertion delay and frequency?
- Q.11 Can we build clock tree without giving skew and insertion delay constraints?
- Q.12 Why via sizes are increasing in lower technologies?
- Q.13 What are the corners used in previous projects and explain how they work.
- Q.14 Role of following cells in latest technologies: Endcap/Endrow cells, decaps, Core fillers, metal fillers, tie-hi/tie-lo cells, Well tap cells?
- Q.15 If spacing is not recommended in your design, what are the other techniques you perform to avoid CC?
- Q.16 Name the types of ECO in P&R. Please provide a

flow to for the different type of ECO.

- Q.17 If you have 40 corners in your design? Do you need to fix violations in all 40 corners?
- Q.18 what are dpt layers in your design and why will go for two masks for those dpt layers
- Q.19 How to find the reason for congestion in particular region?
- Q.20 If skew is violated, how will you fix it?
- Q.21 Schematic diagram and truth table on Isolation cells and retention flops. And switch cell schematic structure
- Q.22 Fixes to Antenna violations: Layer Hoping (jumper/Bridging) and diode insertion (reversebiased), upsize gate (increase gate area).
- Q.23 What are the inputs for running LEC? What are the issues faced in LEC?
- Q.24 what are the inputs to run STA? faced any issues in extraction?
- Q.25 How power stripes are useful in power planning? What are reliability issues?
- Q.26 what happens if duty cycle is ON period 30% and OFF period 70% instead of 50% duty cycle?
- Q.27 What is difference between high drive strength buffer and low drive strength buffer?
- Q.28 What is the amount of input and output delays that are used? For the 70% 30% approach write down the timing report of the In to Out paths with Data Arrival and Data Required times, for a 100MHz clock and 2ns combinational delay.
- Q.29 Why is the term lockup latch is used explain it by developing a scenario.
- Q.30 What is uniquifying the netlist.
- Q.31 what are the components will you see in timing report
- Q.32 In sdc what is synchronous and asynchronous clock groups
- Q.33 In TSMC 7nm technology which track cells you have used and what is the poly width(gate length)
- Q.34 what is logically exclusive and physically exclusive.
- Q.35 why and in what cases false path is applied ? lly

disable timing arc ?

Q.36 write a tcl script to get the number of AND gates in the fanin cone of D pin of register

Q.37 Zero cycle?? Was talking regarding Input to flop paths..

Q.38 What will be the impact on hold time if your skew is more??

Q.39 Why is that you see new violations after Clock tree synthesis? Do the clocks have to be balanced for the spare cells, explain?

Q.40 when -combo switch is used in create_generated_clock

Q.41 Is Hold dependent on frequency? Explain in detail?

Q.42 How shielding exactly reduces crosstalk?

Q.43 Which is preferred in CTS: clock buffer or clk inverter?

Q.44 Pulse width violations: How is it imp & fixes?

Q.45 How extraction is taking OCV & crosstalk effects? How timing is taking both these effects?

Q.46 Is mix Vt possible in CTS? Why? Why one VT cells are preferred in CTS?

Q.47 Explain challenges in previous design

Q.48 cell delays or net delays which one will be more in lower technologies ? why ?

Q.49 If you and your team are working in new technology(ex: 5nm/3nm), what are the ways to escalate issues

Q.50 How will cell sizing affect dynamic, leakage and short circuit power?

Q.51 Design Frequency will affect the cross talk? If yes... why

Q.52 Which type of violations will get .. due to cross talk

Q.53 What is the cross talk effect between 1MHz design to 1GHz design

Q.54 What will happen will cross talk goes high

Q.55 While comparing 1MHz to 1GHz, which design having more hold violations ? why..?

Q.56 What is the clock gate..? explain with diagram and operation

- Q.57 What are the uses of clock gate. How many types of clock gate will be available in design
- Q.58 Xtalk noise on static signals and glitch effects on static signals.
- Q.59 Timing windows effect on xtalk noise.
- Q.60 What the difference between synthesis and High Fan-out synthesis
- Q.61 Will P & R tool reject synthesis netlist with unconnected Input ports?
- Q.62 How do unconnected (or) floating Input ports impact PD flow?
- Q.63 Explain effects of clock domain crossing?
- Q.64 How do you minimize clock dynamic power ?
- Q.65 If two buffers are added in the common path (timing path) ,then what will be the effect on hold in PTSI soft check errors in LVS
- Q.66 why we need softchecks, ERC checks and PERC checks in LVS
- Q.67 how much cap. And tran values u shud specify in the design
- Q.68 why inverter is preffered in CTS over buffer
- Q.69 If typical is 1 volt , then what will be the corners for worst and best and why ?
- Q.70 No. of levels in CTS , can we specify in the clocktree file
- Q.71 in CTS what is preffered first to fix .. logical drcs , skew ,ID
- Q.72 what are the disadvantages of having good skew.
- Q.73 What is temperature inversion?
- Q.74 How is clock uncertainty calculated in your flow? Based on which all parameters ?
- Q.75 Why is clock skew constraint specified as greater than 0 ps or a specific value (30 - 50 ps) ? Why not it is 0 ?
- Q.76 Why we need to do Cell padding. How it will reduce congestion.
- Q.77 Which clock tree is better, the one with shielding on a single spacing or the one with double spacing?
- Q.78 There is path from the launch clock at 2ns and capture clock at 3ns, where the tool checks the setup

- and hold? Write the command for hold check in PT?
- Q.79 If you have a 1000 macros how will start floorplanning.
- Q.80 How will you know that two clocks are synchronous?
- Q.81 What is ICG and its pros and cons?
- Q.82 What are the checks you do after the placement?
- Q.83 Could you please draw internal structure of cmos Nand, nor, isolation, retention, level shifter, end cap , D cap , tap cells etc....
- Q.84 how clock buffer different from normal buffer
- Q.85 How do you prevent transition violations
- Q.86 inputs for crosstalk analysis?
- Q.87 Where do you find the DRV values? how do you analyse in the timing report?
- Q.88 On what factors cell delay depends on?
- Q.89 What are multicycle paths? How we check Setup and Hold for multicycle path?
- Q.90 What happens if i use type of clock buffer which is bad rise time and bad fall time?
- Q.91 What happens if i use type of clock buffer which is bad rise time and good fall time?
- Q.92 What is difference between physical DRC violations and timing DRC violations?
- Q.93 What is IR drop? How it will effect to design?
- Q.94 What is signal integrity? what happens if design have signal integrity problems?
- Q.95 What is the command you will use for adding stripes?
- Q.96 command for adding physical cells like end cap/ tap/ filler etc...
- Q.97 What we do in placement? What we do if congestion exists ? and regarding timing we do setup analysis after placement...
- Q.98 what are calibre Drc's you have seen which are not shown in Innovus
- Q.99 Why CMOS consumes less power?
- Q.100 draw stick diagram of nand gate
- Q.101 What was delay of the buffer that you used in your project?
- Q.102 What is transition violation? How it effects on

- your design power (In terms of short circuit power)?
- Q.103 What is the reason behind that, if the temperature increases why delay increasing?
- Q.104 What is synthesis and explain why it requires?
- Q.105 What are the multicycle paths? How we check Setup and Hold for multicycle path?
- Q.106 What are the inputs for Prime Time?? About Prime time(STA) analysis and reports generated? How do you resolve this issue reported in prime time ??
- Q.107 Timing window concept?
- Q.108 how will u say constraints u r getting from frontend areok
- Q.109 DW-DS for CLK Non -default Routing? Why?
- Q.110 How to solve Electro migration violations?
- Q.111 How to come up with CTS specs: max delay values & max skew values?
- Q.112 How to decide optimum H & W of core, given Aspect Ratio & utilization? Implications if $H > W$ & vice-versa?
- Q.113 How shielding exactly reduces crosstalk?
- Q.114 How buffer insertion to a victim net reduces crosstalk effects?
- Q.115 What are the pros and cons of up sizing and downsizing a cell?
- Q.116 Switching transitions effects the cross talk ?
- Q.117 what is the purpose of via enclosure
- Q.118 Inverter characteristics? Why PMOS will be on at logic 0, and NMOS will be on at logic 1?
- Q.119 How will you conclude a path as multicycle path??
- Q.120 How to decide the placement was good?
- Q.121 How much congestion metric is good to go
- Q.122 Have you done any scripting for your designs?? About db commands..
- Q.123 What will be the impact on hold time if your skew is more??